



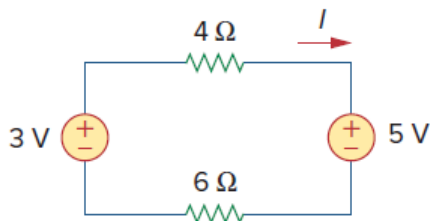
Biju Patnaik University of Technology

CUPGS, Rourkela, Odisha

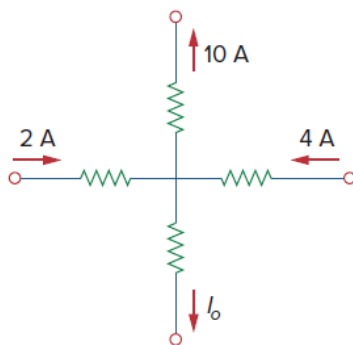
BASIC ELECTRICAL ENGINEERING (BEE)

ASSIGNMENT: 1

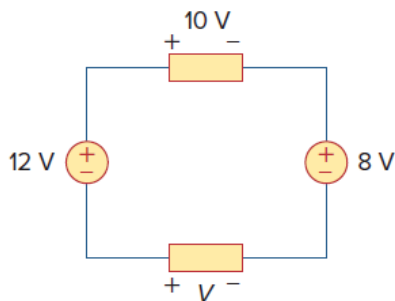
1. The reciprocal of resistance is:
2. An electric heater draws 10 A from a 120-V line. The resistance of the heater is equal to?
3. The voltage drop across a 1.5-kW toaster that draws 12 A of current is:
4. The maximum current that a 2W, 80 k Ω resistor can safely conduct is:
5. A network has 12 branches and 8 independent loops. How many nodes are there in the network?
6. The current I in the circuit of Fig. shown below is:



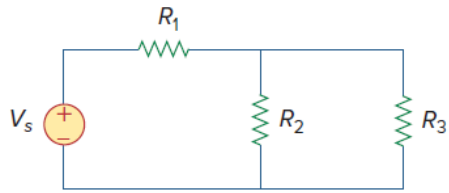
7. State Ohm's law and write its limitations
8. The current I_o of Fig. shown below is:



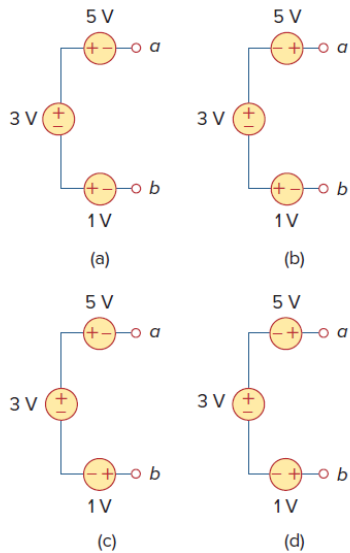
9. In the circuit in Fig. shown below, V is:



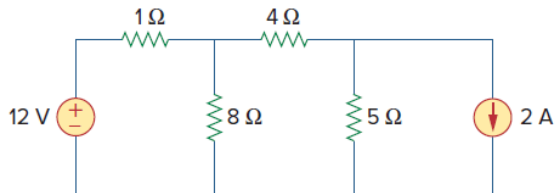
10. In the circuit below, a decrease in R_3 leads to a decrease of, select all that apply:
 (a) current through R_3 (b) voltage across R_3 (c) voltage across R_1 (d) power dissipated in R_2 (e) none of the above



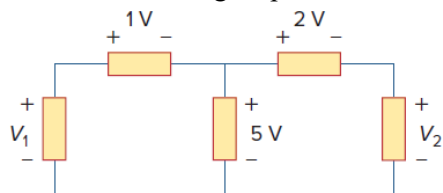
11. Which of the circuits in Figures will give you $V_{ab} = 7 \text{ V}$?



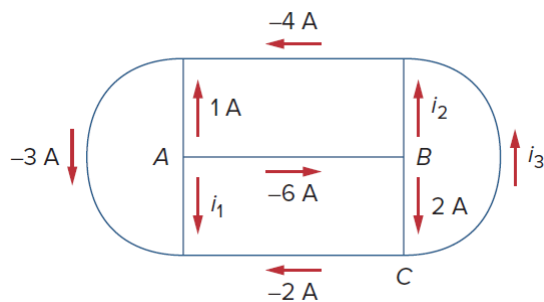
12. Determine the number of branches and nodes in the circuit shown below



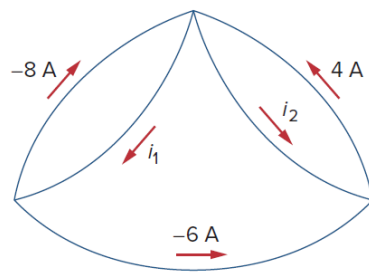
13. In the circuit of Fig. depicted below, calculate V_1 and V_2 .



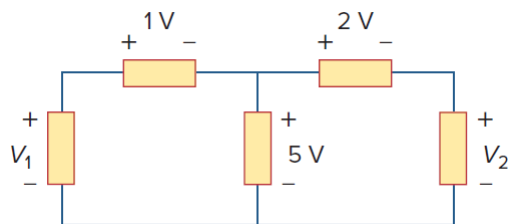
14. Find i_1 , i_2 , and i_3 in Fig. shown below



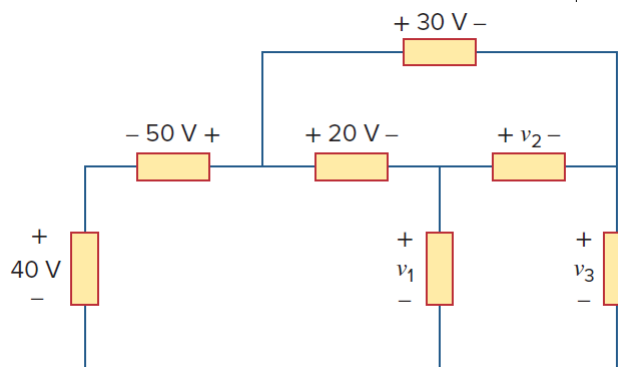
15. Determine i_1 and i_2 in the circuit



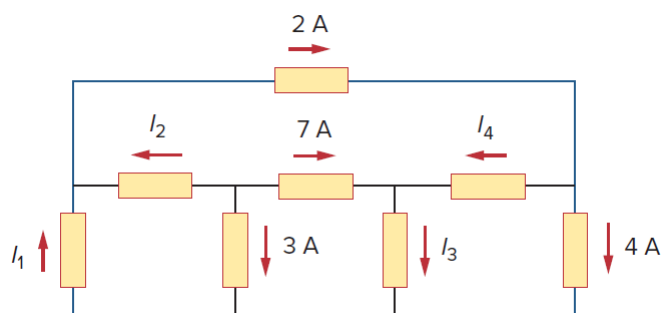
16. In the circuit calculate V_1 and V_2 .



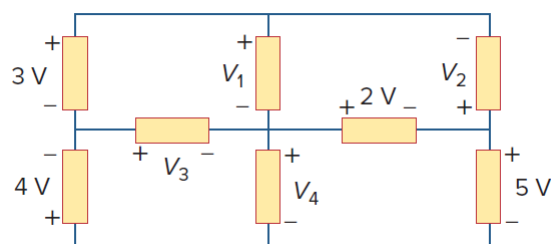
17. In the circuit in Fig. below, obtain v_1 , v_2 , and v_3



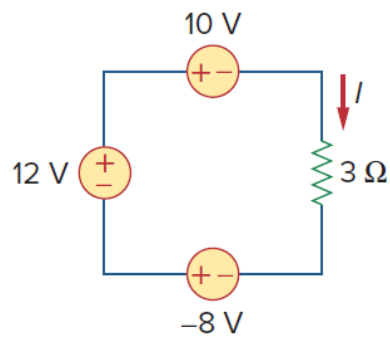
18. For the circuit, use KCL to find the branch currents I_1 to I_4 .



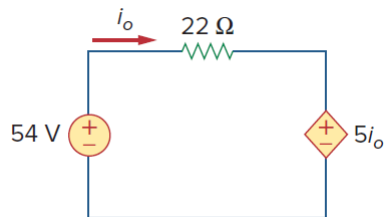
19. Given the circuit, use KVL to find the branch voltages V_1 to V_4 .



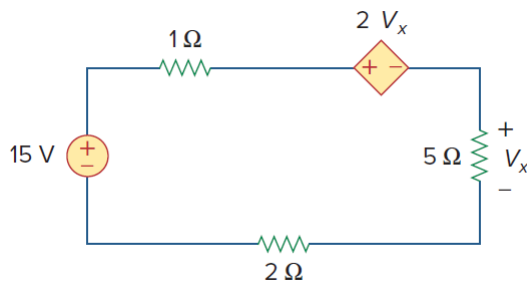
20. From the circuit, find I , the power dissipated by the resistor, and the power supplied by each source.



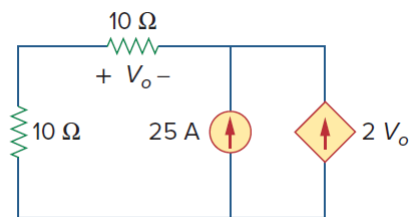
21. Determine i_o in the circuit



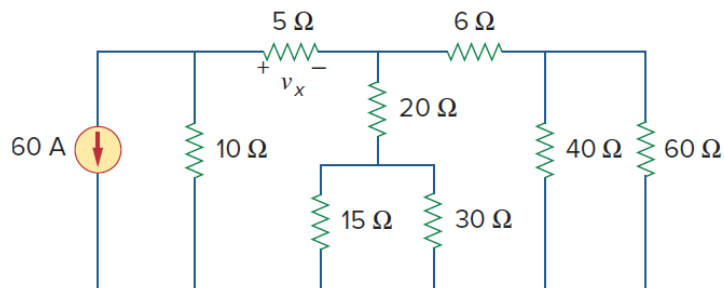
22. Find V_x in the circuit



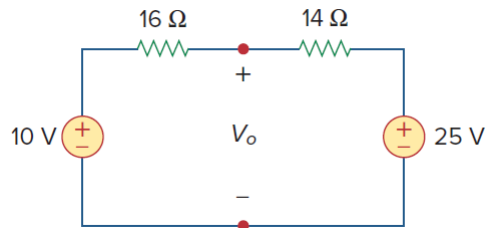
23. Find V_o in the circuit and the power absorbed by the dependent source



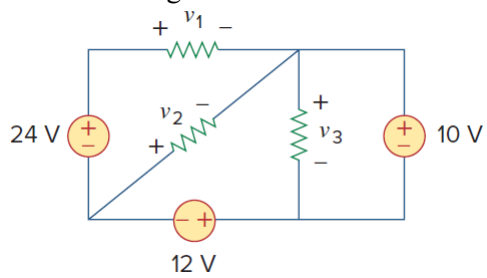
24. In the circuit shown determine V_x and the power absorbed by the 60-Ω resistor.



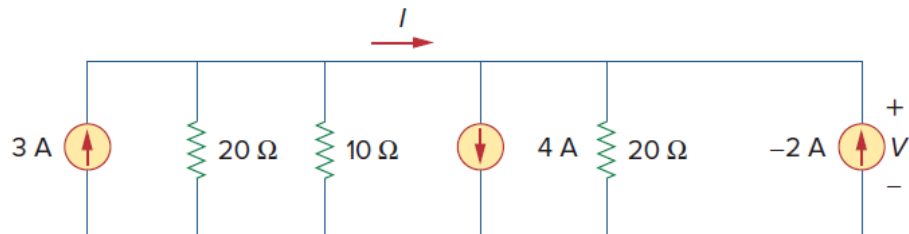
25. Determine V_o in the circuit



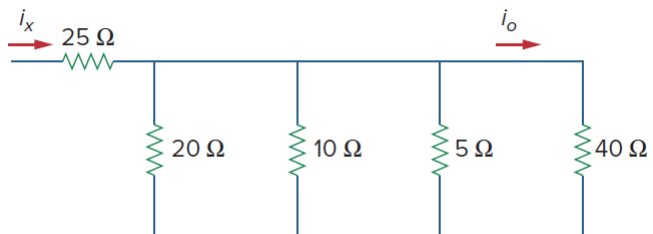
26. Obtain v_1 through v_3 in the circuit



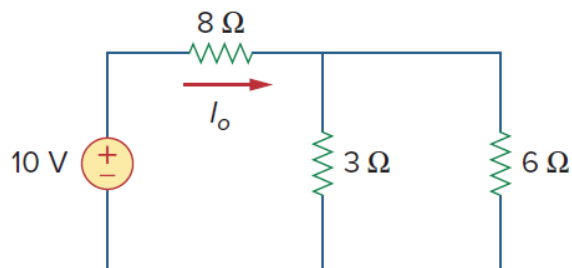
27. Find I and V in the circuit



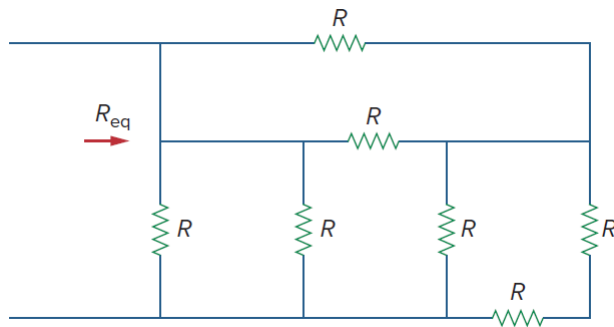
28. For the circuit, $i_o = 3$ A. Calculate i_x and the total power absorbed by the entire circuit.



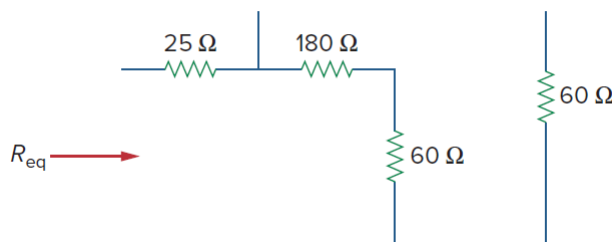
29. Calculate I_o in the circuit



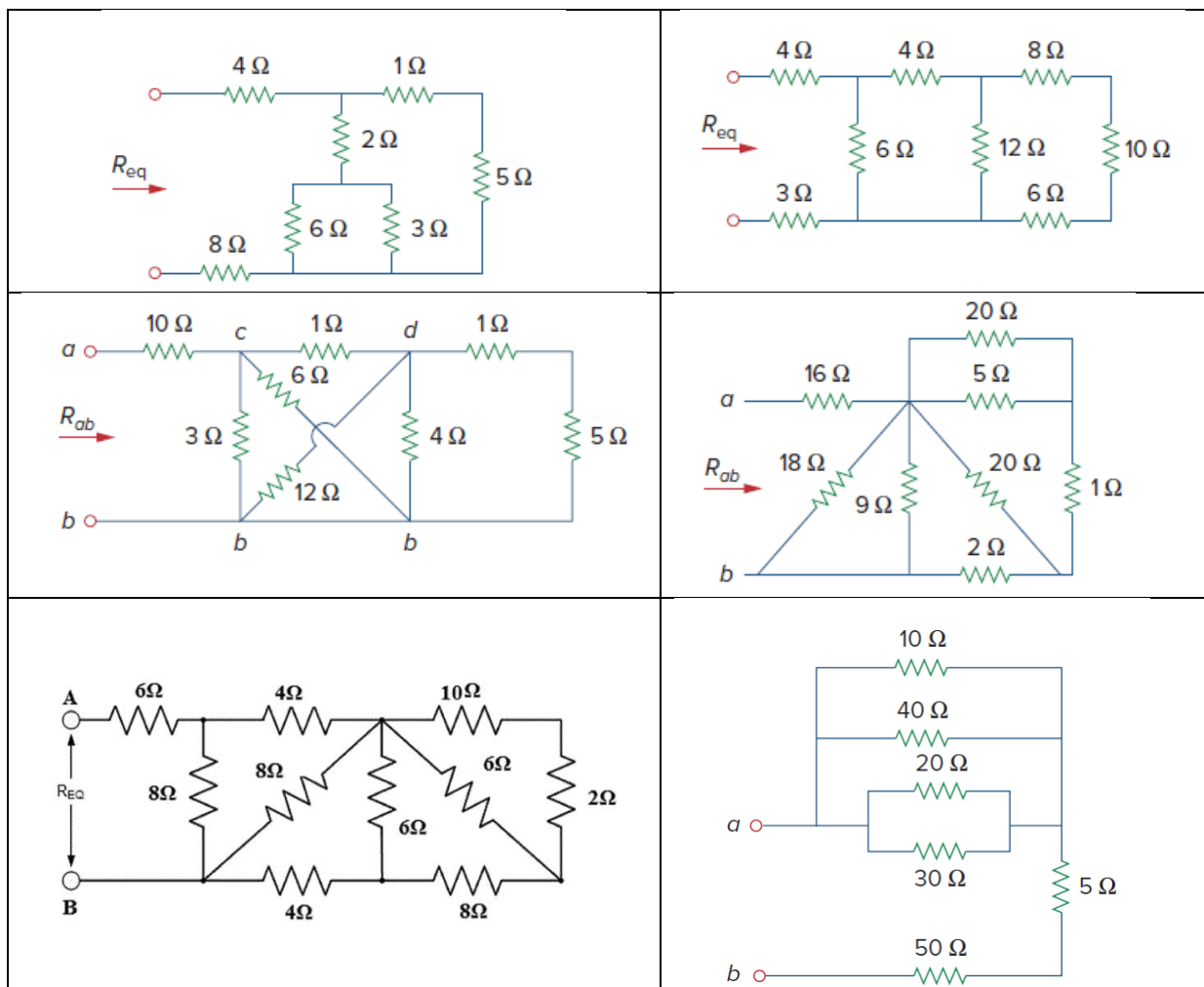
30. All resistors (R) are $10\ \Omega$ each. Find R_{eq} .

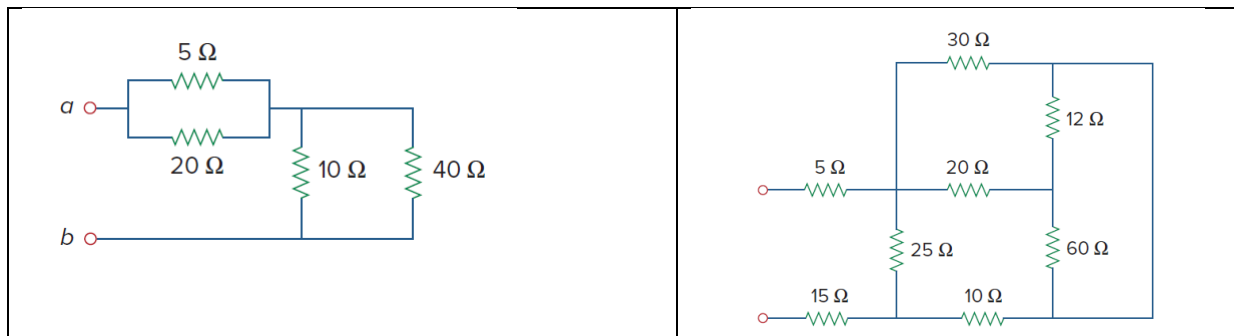


31. Find R_{eq} .

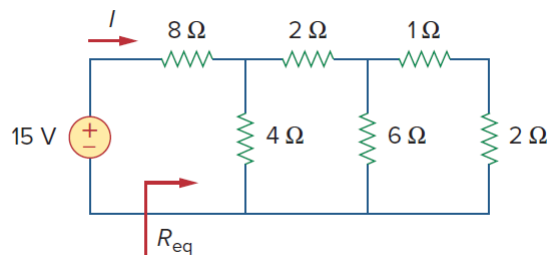


32. Find R_{eq} for the circuits shown

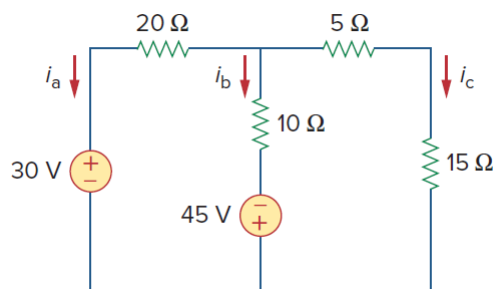




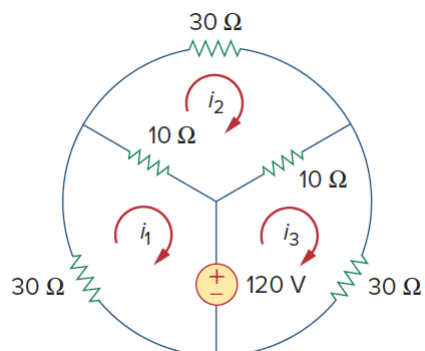
33. For the ladder network, find I and R_{eq} .



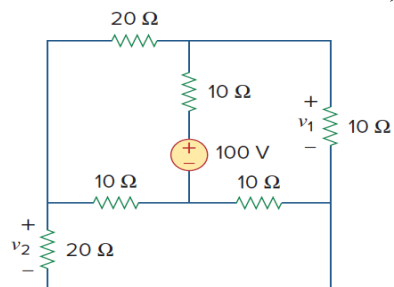
34. Use mesh analysis to obtain i_a , i_b , and i_c in the circuit in



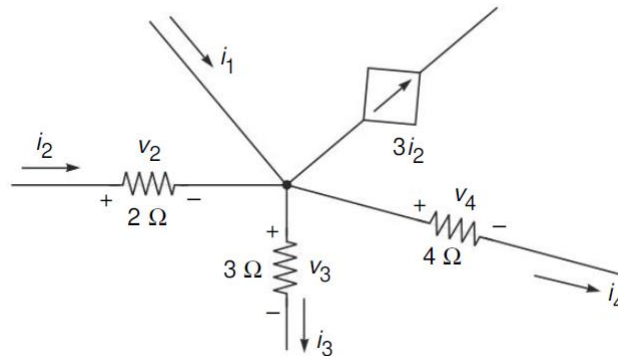
35. Use mesh analysis, Find i_1 , i_2 , and i_3 in the circuit



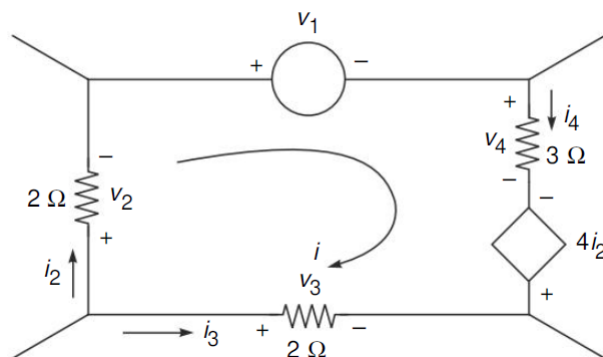
36. Determine v_1 and v_2 in the circuit, (MESH ANALYSIS)



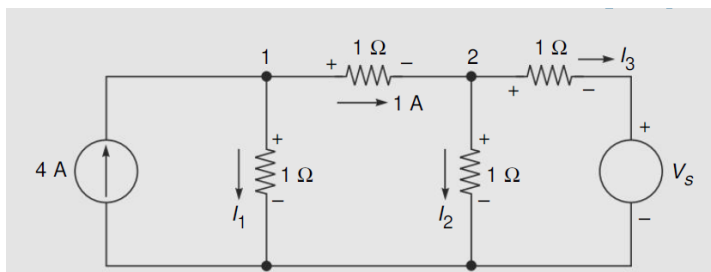
37. Figure below shows one node of an electric circuit. Using KCL, find v_2 . Given: $i_1 = 4\text{ A}$, $v_3 = 3\text{ V}$, $v_4 = 8\text{ V}$



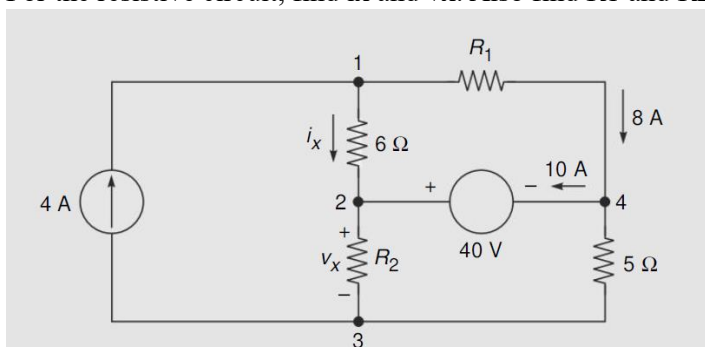
38. Consider the loop of Fig. shown below which forms part of an electric circuit. Find i . Given $v_1 = 6\text{ V}$, $i_2 = 2\text{ A}$, $i_3 = 4\text{ A}$



39. For the resistive circuit shown, find (a) I_1 (b) V_s .



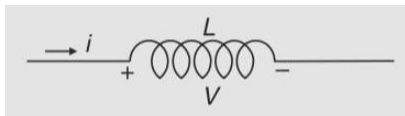
40. For the resistive circuit, find i_x and v_x . Also find R_1 and R_2



41. A resistor $R = 800\text{ Ohm}$ is shown in the figure. Calculate the power absorbed by R at $t = 0.1\text{ s}$ when (a) $i = 50 e^{-10t}\text{ mA}$, (b) $v = 50 \cos 25 t\text{ V}$ (c) $v_i = 10 t^{2.5}\text{ VA}$.

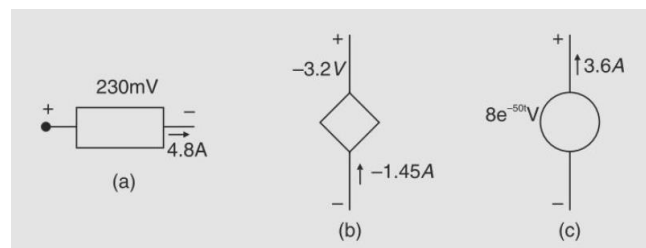


42. The inductor of the figure has $L = 20 \text{ mH}$. (a) Find v at $t = 10 \text{ ms}$ if $i = 8e^{-100t}$.
 (b) Find i at $t = 0.1 \text{ s}$ if $v = 6e^{-12t}$ and $i(0) = 8 \text{ A}$
 (c) If $i = 10(1 - e^{-50t})$ find the power being delivered to the inductor at $t = 40 \text{ ms}$ and the energy stored at $t = 50 \text{ ms}$



43. The current through a capacitor is $i(t) = 9 \sin rt \text{ } \mu\text{A}$. If the energy stored at $t = 200 \text{ ms}$ is $3 \text{ } \mu\text{J}$, what is the value of the capacitor?

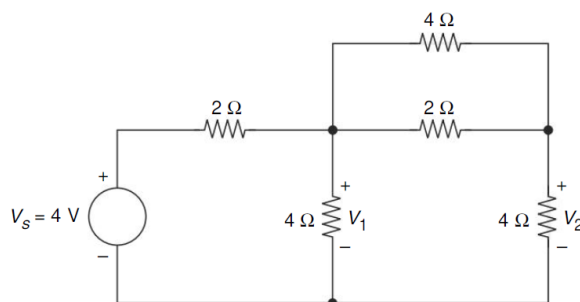
44. (i)



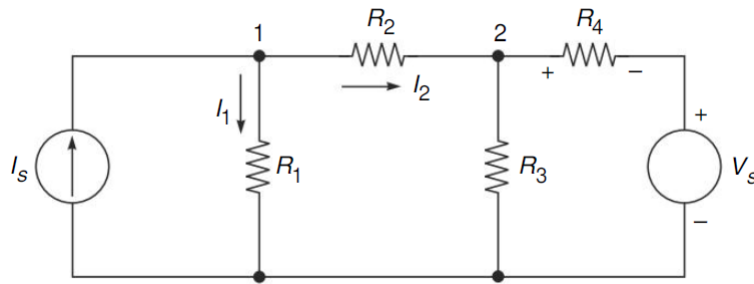
- (a) Is the power being absorbed or generated by the circuit element (a) and what is the value of this power?
 (b) Is the power being absorbed or generated in the circuit element of Fig. (b) and what is the value of the power?
 (c) Find the power being absorbed or supplied by the circuit element of Fig. 1.39(c) at $t = 10 \text{ ms}$.

- (ii) Examine Fig. shown below.

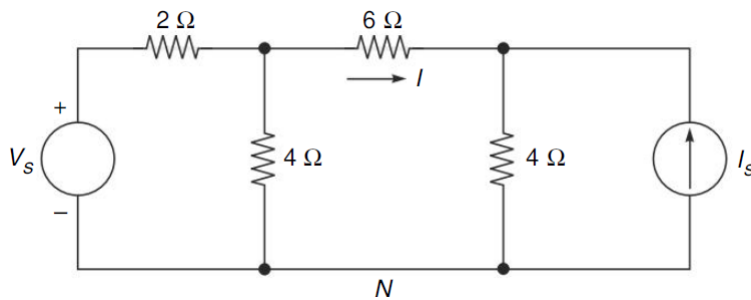
- (a) Express the voltage across each resistor in terms of V_s , V_1 and V_2 .
 (b) Write the necessary KCL equation at the nodes in terms of V_s , V_1 and V_2 .
 (c) Solve for V_1 and V_2 .



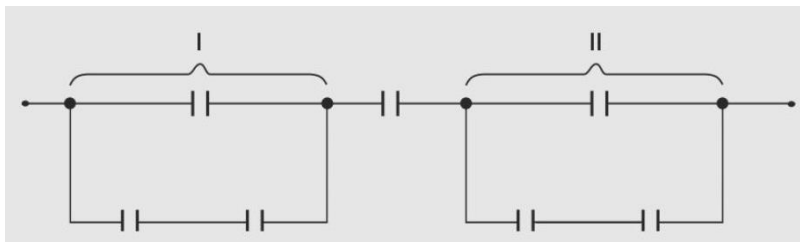
45. For the resistive circuit shown in Fig. below, $I_s = 4\text{ A}$, $I_2 = 1\text{ A}$, $R_1 = R_2 = R_3 = R_4 = 1\text{ Ohm}$.



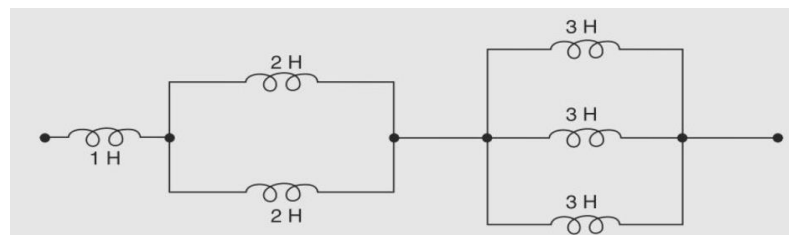
46. For the circuit below. (a) Given $V_s = 12\text{ V}$, determine I_s so that $I = 0$. (b) Given $V_s = 14\text{ V}$, and $I = 1\text{ A}$, find I_s .



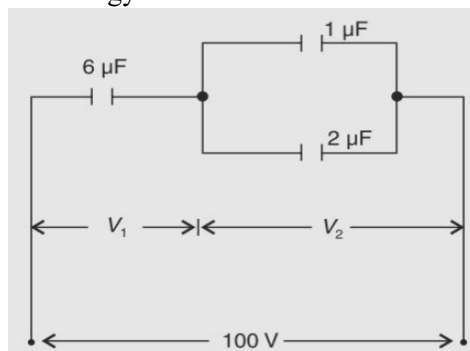
47. For the capacitor circuit determine C_{eq} if each capacitor is $10\text{ }\mu\text{F}$.



48. Find Leq for the inductor circuit



49. For the capacitor circuit, determine (a) the charge on each capacitor (b) the stored energy in each capacitor and total stored energy.



50. Two inductors when connected in series have an inductance of 4H and $\frac{3}{4}\text{H}$ when connected in parallel. Determine the inductance of each inductor